

Senior Research Experience

Recreational Marijuana and State-Level Employment

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Abstract

In this report, I will investigate whether or not the legalisation of recreational marijuana leads to a significant and measurable effect on state level employment within the United States. To do so, I will implement a staggered, state-year fixed effects DiD model using state-year panel data. The data includes all 50 states, years from 2000-2022, and several control variables that cover primarily demographic and economic factors. The model will follow the outline of the Sun and Abraham (2021) model, and will then be continuously developed to account for different concerns with the data. These developments include a double interaction model, a model involving inverse probability weighting, and an alternative response variable.

Across all variations of our models, we see no significant effects immediately following legalisation, but approximately 7 years after legalisation we do see evidence of a positive relationship. These results suggest that the benefits of passing recreational marijuana legalisation legislature on employment levels are real yet delayed, and are likely as a result of the time required for the market to adjust to new infrastructure, licenses to be obtained, and similar such economic-based activities.

Introduction

In recent US history, the issue of whether or not to legalise recreational marijuana has gained major traction and many states have implemented policy that up-ends prior strict restrictions on the drug. The pace has only increased with more and more states passing such legislature in the past couple of years. With this “Green Wave” comes a plethora of concerns, but a lot of them surround the economy and labour markets. Those in favour of legalisation cite arguments like the creation of new industry / further development of existing industry and new avenues for tax revenue (as well as other, less economy-specific ideas such as ease of regulating product). On the other side of the argument, skeptics argue a decrease in labour productivity which would lead to a series of negative side-effects within the US labour market. Both sides of the debate raise valid claims, and it is my goal to try to answer them as best as I can.

More specifically, I aim to answer the question of whether legalising recreational marijuana has an impact on USA state employment levels, if so, how immediate is that effect, and in what direction does it trend? It is my hypothesis that while effects will not be seen immediately, they should be expected within the first 3-5 years as a lot of the industry exists already, it is just unaccounted for due to its illegal status pre-legalisation. That being said, in order to expand from previous studies, I do not plan to restrict the post-legalisation period (in most prior works, it IS restricted). This should allow me to truly test the validity of my assumptions. Furthermore, I rationalise that after a given amount of time, the treatment effect should wane and regress back to being insignificant. In order to best study this, I use a staggered DiD approach to explicitly test for how long post-legalisation the treatment effect (if any) is experienced.

Lit Review

This research is, of course, not the first of its kind. It is grounded in prior research and build upon previously discovered results to further the field of study. The legalisation of marijuana recreationally led to many conflicts for policy making, and a lot of the research was made subsequently. Scholars largely agree that legalisation does produce measurable effects on employment and income, however they disagree upon the direction and magnitude of those effects. There are 3 key works that truly embody this uncertainty:

The first is that of Sara Varon, titled “The Green Wave: Effects of Legal Marijuana on The Labor Markets”. She focuses specifically on industries that are directly impacted by marijuana, and finds that legalisation is related to an increase in both wage and employment in those relevant industries. Her reasoning was that new jobs are being created within those industries that previously did not exist. She goes on to say that by making the drug legal, the market does not magically appear but rather becomes regulated, resulting in labour and capital into a taxable and recordable industry.

The second article is “The Impact of Recreational Cannabis Legalization on Cannabis Use and Associated Outcomes: A Systematic Review” by James MacKillop. MacKillop is less eager to discern significant results and finds that the effects of legalising marijuana recreationally are widely inconsistent due to differences in populations, outcome measures, and natures of different studies. It is of note that MacKillop’s article is not so much a representation of his own study, but rather an analysis of a compilation of many other previously conducted studies. His conclusion is that the field of study is so convoluted due to 3 main issues: incomparable methodologies, differences in state definitions of legalisation, and a lack of time post-treatment which makes it hard to make causal claims with significance.

The third and final article that inspired my research is “Cannabis Use, Employment, and Income: Fixed-effects Analysis of Panel Data” by Ioana Popovici and Michael T French, in which negative trends are observed. This research uses a fixed effects model that tries to account for individuality and effects on the individual rather than a holistic approach like Varon. Popovici and French find that individuals who smoke marijuana after legalisation policy is passed see a decrease in probability of being hired - a decrease in hireability essentially. In other words, they find that the same policies that Varon found to have a positive impact on the labour market simultaneously cause individuals to increase personal consumption and henceforth experience negative side effects in the job market.

Looking at these three articles together, it is clear that the effect of legalising marijuana is highly dependent on what sectors the researcher is looking into. That being said, both Varon and Popovici and French’s studies look deep into specific aspects of the policy’s aftermath. Therefore, my focus will be trying to use what they learnt about their specifications and apply to a more general overview of employment with the goal of testing personally if there is an effect or if the uncertainty seen by MacKillop still exists after more time has passed.

The Data

In order to carry out this study, several variables were gathered and compiled into one main dataset. At its largest, this dataset covered years from 1974-2022 with controls for economic and demographic factors for all 50 states and some territories (and the District of Columbia). This data comes from sources such as the Bureau of Economic Analysis (BEA), FRED, and StatsAmerica.

In order to reduce computational cost, and to ensure all data used was relevant this dataset was severely cut down. The final result was state-year panel data spanning only the 50 states, and only from the year 2000-2022. It contains the following variables for any given state during any given year:

- Number of People Employed (The Outcome)
- GDP
- Median Household Income

- Total Population
- Percent of Population that is Male
- Percent of Population that is Black

As a result, the dataset contains 1,150 observations prior to restrictions made for each model.

In order to best conduct my study, transformations were applied uniformly to every model. The employment level data was transformed using the natural log such that variance across states that differ vastly in size is stabilised, but also allows the DiD coefficients to be interpreted as percent changes which is a more important measure of analysis in this regard (as compared to raw numbers). GDP and Median Income undergo a natural log transformation for the same reason. These two variables depend heavily on time, so taking the natural log helps to better fit them to a linear relationship with the outcome (thus increasing interpret-ability and significance further). Percent Black and Percent Male are standardised using z-scores. We cannot take the log of these as they are already percentages, however we need to be able to better account for the time-dependent variance that arises from these variables.

What is Difference-in-Differences?

Difference-in-Differences is a model used by economists when attempting to identify causal effects of some treatment between a treatment group and a control group. These studies are usually used when a natural experiment arises, such as a specific policy is passed into law or when a major event (like an oil spill) occurs. The goal of the study is to identify differences between the two groups over time as a result of the event. The treatment group is the group that experiences or is affected by the event, and the control group is a group comprised of similar individuals/regions. The two groups have to be similar before the treatment event such that we are able to assume the treated group would continue to be similar to the control group if the treatment event had never occurred. This assumption is known as the parallel trends assumption. Economists graph the observed differences and calculate the difference between what is actually observed to the assumed counterfactual (what would have happened without the treatment event).

In the case of this study, we are also implementing a DiD methodology. Specifically, we have 24 US states that legalised marijuana recreationally at some point between 2012 and 2022 and 26 US states that did not legalise recreationally. The states that did pass legislation making recreational marijuana legal for consumption will be our treated group, and the states without such legislation will be our control group. Below is a list of each group with the exact year of treatment for the treated states:

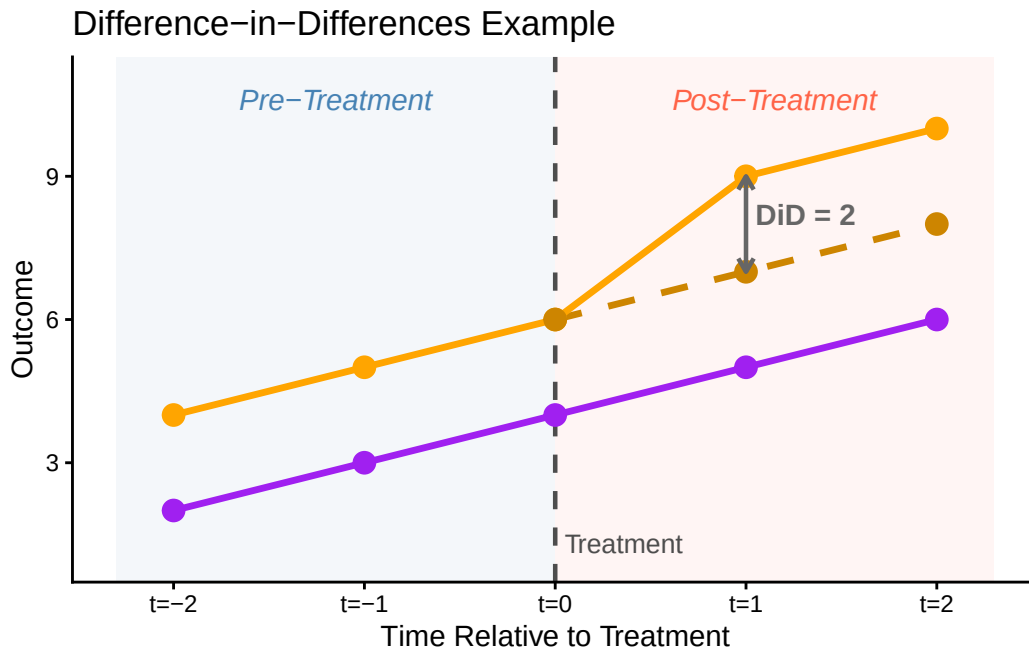
- Treated Group: Alaska (2014), Arizona (2020), California (2016), Colorado (2012), Connecticut (2021), Delaware (2023), Illinois (2019), Maine (2016), Maryland (2022), Massachusetts (2016), Michigan (2018), Minnesota (2023), Missouri (2022), Montana (2020), Nevada (2016), New Jersey (2020), New Mexico (2021), New York (2021), Ohio

(2023), Oregon (2014), Rhode Island (2022), Vermont (2020), Virginia (2021), Washington (2012)

- Control Group: Alabama, Arkansas, Florida, Georgia, Hawaii, Idaho, Indiana, Iowa, Kansas, Kentucky, Louisiana, Mississippi, Nebraska, New Hampshire, North Carolina, North Dakota, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, West Virginia, Wisconsin, Wyoming

Now that we have defined our groups, we now need to define a treatment event. In our study, the treatment event is the passing of legislation that makes recreational marijuana legal, however since there are many years across which such laws are passed, we will have to use what is known as a Staggered DiD approach. All this means is that we have a continuous, or rolling, treatment event, and our model will have to account for that.

The following graph illustrates the core logic of a Difference-in-Differences model that we just discussed. Before the treatment event occurs ($t < 0$), both the control and treatment groups can be observed to have parallel trends. At $t = 0$, the treatment occurs. The DiD estimate is the gap between the treatment group's actual outcome and the counterfactual (represented by the dashed line). The counterfactual is determined by assuming the treated group would continue to behave like the control group.



You might have noticed that this graph only shows one treatment year. That is a logical belief, however this graph is demonstrating the core idea of a staggered DiD approach: relative time. Instead of using specific calendar years, we replace actual years of treatment with a counter centered around the treatment event, in which the year of legalisation is equal to $t=0$, the year

before is equal to $t=-1$, the year after legalisation becomes $t=+1$, and so on. This allows us to compare treatment events that occurred in different years on a standardised timeline, as if the treatment events had happened at the same time.

Preface: Model Contents

Besides the variable transformations, two more specifications will be applied to all models used within this report.

The first specification is the use of the Sun and Abraham (2021) model to allow for staggered and valid treatment effect results. Normal TWFE Fixed Effects models do not properly account for what happens when the treatment has varying magnitudes of effect in different places. The SunAb model is a perfect solution to this problem due to one main underlying assumption about the two options: In the typical TWFE model, states that just underwent treatment could be compared to states that underwent treatment years prior. This is NOT a valid comparison in the context of DiD. In the SunAb model, only states that have not YET undergone treatment can be used as control group comparisons. In other words, every state starts out in the control group, and as they become treated, they leave the control group and can never be used as part of that group again. The SunAb model also uses relative time to make comparisons which further adds to its robustness for this type of DiD study.

The second specification is a more complex one: clustering. In short, it is a type of unsupervised machine learning algorithm that groups similar observations together. Unsupervised just means that this process occurs without need for human intervention at each step. In this study, I use clustering to group observations with equivalent state titles. This accounts for the idea that each state has its own set of unobservable factors, giving us more robust results. Failing to account for these state-specific trends would cause standard errors to be artificially small, making results appear more statistically significant than they actually are. Given that our treatment variable — legalisation status — also varies at the state level, clustering by state is a valid decision.

Model 1: State-Year Fixed Effects Staggered DiD

The first model uses a staggered Difference-in-Differences approach with a single post-legalisation interaction term. State and Year Fixed Effects are included and as a result absorb the effects of the Treat and Post terms when observed individually due to their time-invariant differences between states. In other words, they are binary indicators, so after legalisation they will not change. We use the control variables: $\log(\text{GDP})$, $\log(\text{Median Income})$, total population, % male, and % Black. These are represented by the ΓX term.

$$\log(\text{Employ}_{st}) = \beta_0 + \beta_1(\text{Treat}_s \times \text{Post}_{st}) + \Gamma X_{st} + \alpha_s + \lambda_t + \varepsilon_{st}$$

where:

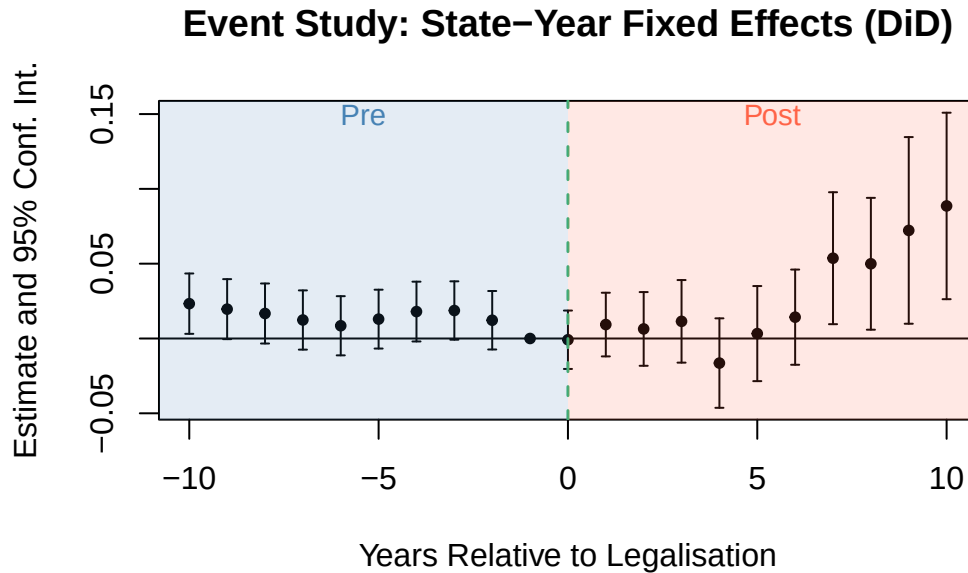
$$\Gamma X_{st} = \gamma_1 \log(\text{GDP}_{st}) + \gamma_2 \log(\text{Inc}_{st}) + \gamma_3 \text{Pop}_{st} + \gamma_4 \% \text{Male}_{st} + \gamma_5 \% \text{Black}_{st}$$

Table 1: State-Year Fixed Effects

	Model 1:
Post	0.0009 (0.0140)
$\log(\text{GDP})$	0.0662* (0.0268)
$\log(\text{Median Income})$	0.0088 (0.0091)
Population (Millions)	0.0246*** (0.0060)
Pct Male (z)	-0.0326 (0.0242)
Pct Black (z)	0.0873 (0.0672)
Num.Obs.	930
R2	0.999
R2 Adj.	0.999
R2 Within	0.296
R2 Within Adj.	0.291
Std.Errors	by: State
FE: State	X
FE: Year	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

We see that our main variable of interest from Model 1 - “Post” - is lacking in both magnitude and statistical significance. This indicates a lack of treatment effect, while our GDP and Population controls do show significance. This is telling us that the model is reacting primarily to size of state (bigger states tend to have more people and higher GDP). Before we adjust these though, it is worth looking into the year-by-year effects using an event study graph.



This event study graph appears to tell a more holistic story as compared to the regression output table. We see that the pre-legalisation trends align with our hopes: the standard errors are clustered around 0 which satisfies the parallel assumptions trend as it means both control and treatment states followed a similar trend before legalising.

On the post-legalisation section of the graph, we gain further insight that was missing from the table. The table told a story of no effect whatsoever, however it appears that was a result of averaging the effect for each year: years 1-6 post-legalisation show no significant treatment effect, however at years 7-10 post-legalisation there does appear to be a slight upward trajectory. The confidence intervals do widen though as a result of fewer observations thus higher levels of uncertainty.

Combining what we learnt from both the table and event study, model 1 is suggesting that if a treatment effect does exist, it will not occur immediately. We will need to account for this as we continue our modelling.

Model 2: Double Interaction State-Year Fixed Effects Staggered DiD

As aforementioned, the goal of model 2 will be to aim to resolve the questions left unanswered by model 1: if the true effect is zero in early years post-legalisation and positive in later years, the two will average out and the aggregated estimate will be attenuated toward zero, thus giving misleading results. Model 2 will address this by creating two post-legalisation periods. The first will be called “During” and this will represent the period between legalisation and (legalisation + n-years). The second period will be called “Post” and will represent the period after (legalisation + n-years). This segmented approach will allow me to test whether there is actually evidence of a treatment effect in later years.

In order to maintain robustness, I will iterate through a series of model options in which the length of time that “During” covers changes. In other words, I will run a model for each different length of period spanning from n=3 to n=8 (given “During” = (legalisation + n-years)). The model that shows the earliest sign of a treatment effect will be the model best suited to making strong conclusions.

It is also worth noting that the Sun and Abraham event study implicitly captures this type of specification without requiring a manually defined threshold. This is because the model produces year-by-year coefficients, the onus falls upon the reader to make the observation that at some given year, the coefficients give evidence of a treatment effect. It is due to this underlying current that the literal model with defined period lengths and the event study complement one another in the sense that the event study visually demonstrates the shape of the effect, while the regression mathematically proves the theory behind the event study.

So; model 2 replaces the single post indicator with two interaction terms that distinguish between the “during” period and the “post” period. The controls remain unchanged.

$$\log(\text{Employ}_{st}) = \beta_0 + \beta_1(\text{Treat} \times \text{During})_{st} + \beta_2(\text{Treat} \times \text{Post})_{st} + \Gamma X_{st} + \alpha_s + \lambda_t + \varepsilon_{st}$$

Let’s take a look at the regression outputs:

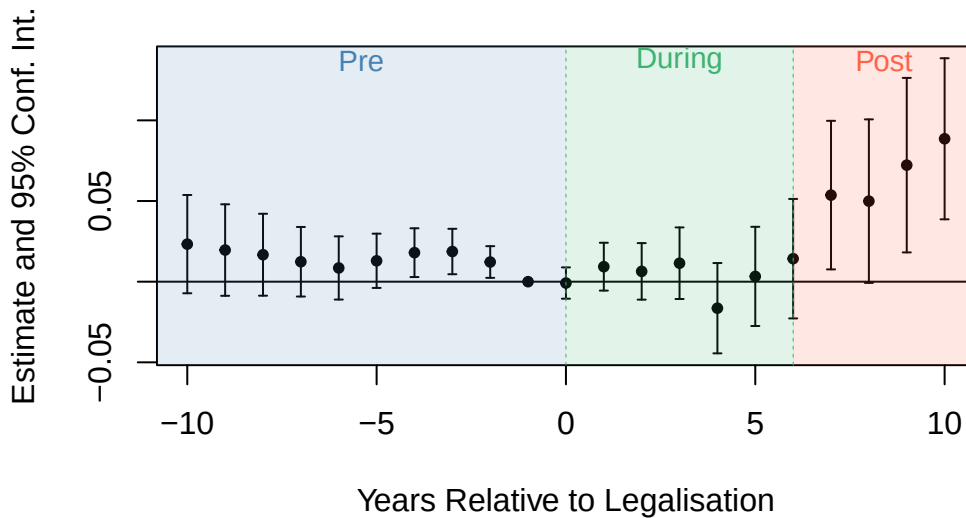
Table 2: Comparison of Treatment Windows

	T+3	T+4	T+5	T+6	T+7	T+8
During	−0.0053 (0.0121)	−0.0059 (0.0123)	−0.0049 (0.0127)	−0.0033 (0.0131)	−0.0015 (0.0134)	−0.0002 (0.0137)
Post	0.0170 (0.0226)	0.0295 (0.0227)	0.0424+ (0.0224)	0.0625** (0.0210)	0.0656** (0.0213)	0.0749*** (0.0197)
log(GDP)	0.0644* (0.0267)	0.0651* (0.0265)	0.0653* (0.0267)	0.0640* (0.0266)	0.0636* (0.0266)	0.0637* (0.0266)
log(Median Income)	0.0086 (0.0090)	0.0085 (0.0089)	0.0084 (0.0088)	0.0087 (0.0089)	0.0086 (0.0088)	0.0087 (0.0090)
Population (Millions)	0.0244*** (0.0058)	0.0242*** (0.0057)	0.0242*** (0.0056)	0.0241*** (0.0055)	0.0242*** (0.0057)	0.0242*** (0.0058)
Pct Male (z)	−0.0324 (0.0240)	−0.0329 (0.0240)	−0.0336 (0.0241)	−0.0347 (0.0244)	−0.0339 (0.0244)	−0.0336 (0.0244)
Pct Black (z)	0.0894 (0.0665)	0.0902 (0.0660)	0.0924 (0.0661)	0.0965 (0.0668)	0.0935 (0.0669)	0.0897 (0.0670)
Num.Obs.	930	930	930	930	930	930
R2	0.999	0.999	0.999	0.999	0.999	0.999
R2 Adj.	0.999	0.999	0.999	0.999	0.999	0.999
R2 Within	0.304	0.314	0.321	0.328	0.320	0.312
R2 Within Adj.	0.298	0.308	0.316	0.323	0.314	0.306
Std.Errors	by: State	by: State	by: State	by: State	by: State	by: State
FE: State	X	X	X	X	X	X
FE: Year	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The results for model 2's table output are consistent with our assumptions from model 1: the “During” coefficient is lacking in magnitude and statistical significance for all thresholds, whereas the “Post” coefficient becomes significant when “During” = (Legalisation + 6 Years), indicating the presence of what can be referred to as a development period: the time it takes for the marijuana industry to properly develop enough to have an impact on the labour market. Thus, the output is in favour of a delayed treatment effect.

Event Study: Two Interaction Model



Model 2's event study further solidifies the idea of a delayed treatment effect. We see that the "During" period mirrors similar trends to the pre-legalisation period, whereas the "Post" period ($n > 6$) sees a slight upward trend. Once again, the confidence intervals grow in size due to fewer observations for these years.

Overall, model 2 both mathematically and visually confirms that there is evidence of a delayed treatment effect or in other words, demonstrates a likelihood of there being a development period for the marijuana industry.

Propensity Scores and Inverse Probability Weights

In Difference-In-Differences analysis, one of the main assumptions required to be believed to be true is the parallel trends assumption, meaning we need to have reason to suspect the treated group would be observed to experience the same trend as the control group in the event that the treatment never occurred. While our current event study models do depict evidence that this assumption is satisfied, it is worth ensuring that this really is the case. As such, I will assign weights to each of the states that will then impact the way the pre-legalisation period groups are formed.

Before diving into the math and formulas, it is important to understand why we will do this, and it is because states naturally differ from one another. No two states are identical, even if they may appear similar across several different variables. As such, we need to "re-weigh" the pre-treatment group observations such that the control states are more representative of the

treated group. We know this is needed due to the significance of variables like Population in our regression outputs.

The weighting itself is broken down into two stages: propensity score creation and weight assignment. This is how it works in a step-by-step breakdown:

1. Calculate the average for each of the control variables for any given year across all treated states.
2. Create a logistic regression model using our control variables that produces a probability of a state being treated. This output is a score on a 0-1 scale.
3. Insert the score from the logistic regression model into either $1/p$ or $1/1-p$, where p = score for any given state. Use $1/p$ when the state in question is actually treated in reality, and use $1/1-p$ if the state in question never undergoes treatment.

This feels complicated, but the underlying logic is much simpler than you might expect: rather than treating all 26 control states as perfectly valid comparisons for the 24 treated states, reweigh them all based on their similarity to the average state that does legalise. As a result, control group states that are predicted to get treated receive a higher weight than control group states that are not predicted to legalise. This just means that states that look most like the average treatment state are given higher importance in our model's pre-period.

Why does this matter? Well, because the control group is more closely resembling the treated group, the parallel trends assumption is more defensible due to increased robustness. It also reduces the risk that the estimates seen thus far have been tainted by pre-existing differences in trends between the treated and control groups, and reduces the selection bias in the DiD estimates.

What does this actually look like? Here is the logistic regression model:

$$\Pr(\text{Treated}_s = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \overline{\text{Pop}}_s + \beta_2 \overline{\text{Inc}}_s + \beta_3 \overline{\text{Male}}_s + \beta_4 \overline{\text{Black}}_s)$$

And here are the weight equations:

$$w_s = \begin{cases} \frac{1}{\hat{p}_s} & \text{if Treated} \\ \frac{1}{1 - \hat{p}_s} & \text{if Control} \end{cases}$$

where $\hat{p}_s = \Pr(\text{Treated}_s = 1)$ is the estimated propensity score for state s .

Model 3: Weighted State-Year Fixed Effects Staggered DiD

Now that we have defined our weights, we need to create another Sun and Abraham model that best reflects what we are attempting to do. Luckily, contained within the `sunab` function, there is an option to specify weights. We do so by calling on the weights that we have assigned to each state within our dataset. This means the underlying model is syntactically exactly the same as previously seen, besides the applied weighting to the pre-legalisation period.

It should be noted that the treated group is also assigned weights, and these act in the inverse way that the control group's weights do. For the control group, high propensity scores lead to high weights, meaning more influence on the results. For the treated group, high scores lead to lower weights in order to not exaggerate the importance of already likely legalisation states. Instead, it gives the states that did legalise, but were predicted to not legalise, a larger influence. We do so as we assume there is some unobserved rationale behind the situation, and we need to find some way to try to account for those unobservables.

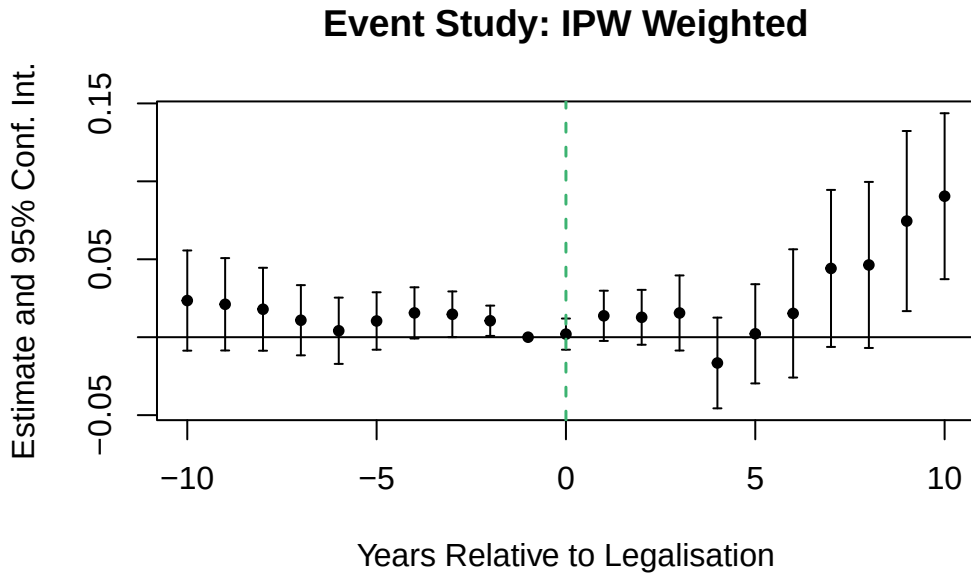
This rebalancing of our pre-legalisation trends does slightly change how we view the model though. Before, when analysing the outputs, we were asking if states that legalised recreational marijuana outperformed all control states with regards to employment levels. Now, we are asking if states that legalised outperformed the control states that were most similar to them. This model can be improved by adding more variables to the logistic regression model such as terrain type, region, average temperature, majority political sway, etc. with the goal of best capturing the characteristics of the state as possible. That being said, I believe this simpler model to be perfectly adequate given they are the major identifier variables in terms of comparing states to one another.

Enough is enough though, let's get into the results a little more. I have run the weights on both model 1 (classic DiD) and model 2 (where $\text{during} = t > 6$) in order to make a comparison and doubly confirm our theory of delayed treatment effects.

Table 3: IPW Weighted Comparison

	Post	Post (T>6)
Post	0.0056 (0.0138)	0.0574* (0.0221)
During		0.0015 (0.0133)
log(GDP)	0.0692** (0.0227)	0.0686** (0.0223)
log(Median Income)	0.0032 (0.0035)	0.0032 (0.0034)
Population (Millions)	0.0246*** (0.0055)	0.0241*** (0.0050)
Pct Male (z)	-0.0400+ (0.0207)	-0.0421* (0.0207)
Pct Black (z)	0.0896 (0.0738)	0.0995 (0.0740)
Num.Obs.	930	930
R2	0.999	0.999
R2 Adj.	0.999	0.999
R2 Within	0.321	0.345
R2 Within Adj.	0.317	0.340
Std.Errors	by: State	by: State
FE: State	X	X
FE: Year	X	X
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		

In Table 3, we see that for model 1, we still see no treatment effect whatsoever. On the other hand, we still see an effect after relative year 6 in model 2's output, and no effect for the "During" period, which aligns perfectly with our theory. While validating our theory, the weighted results also inform us that the observed delayed effects are not simply due to poor pre-legalisation parallel trends, but actually due to the treatment itself.



The event study seen above only serves to further prove our argument of a delayed effect. We maintain valid parallel trends pre-treatment, and then see a clear increasing trend in employment from year 4-10, with true effects being witnessed around relative year 9. This tells us that SOME of the treatment effects previously seen may have been down to poor pre-treatment group comparisons, but given we maintain an effect, this is okay.

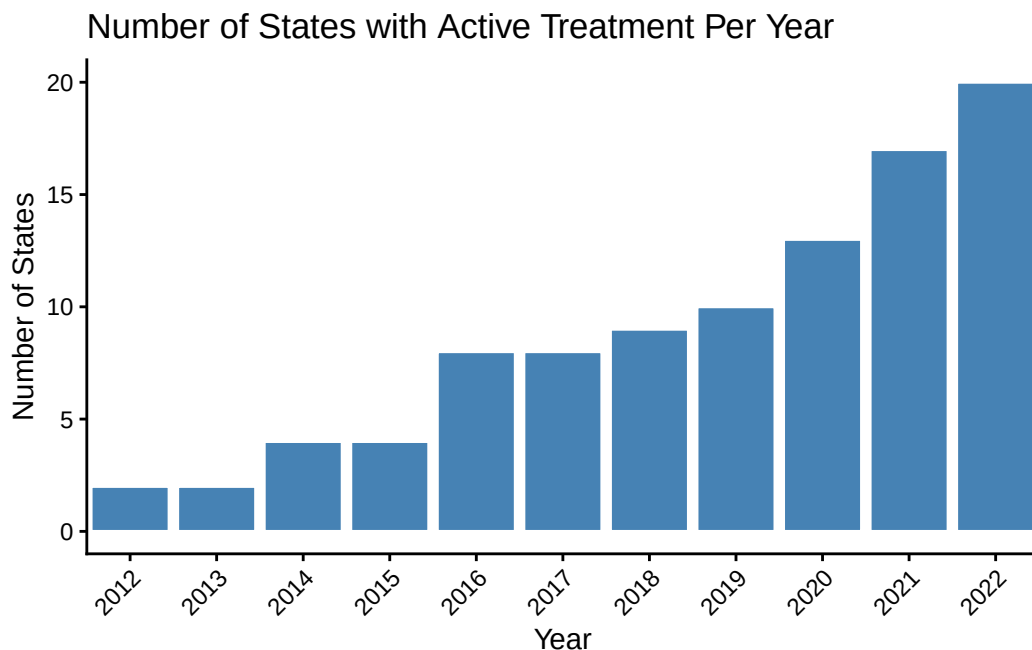
States with Active Treatment Count

Before creating my final model, it is important to first look into a key aspect of the data that has been somewhat neglected for the duration of this research. I must construct a visualisation that tells me exactly how many states have an active treatment for any given year. What do I mean by this? Here is an example:

- Let's say California legalises in 2016. California's active treatment spans from 2016-present. On the other hand, if New York doesn't legalise until 2021, then New York's active treatment period is from 2021-present. If these were our only two observations, then the visualisation should show that the number of active treatments is equal to 0 from 2013-2016, equal to 1 between 2016-2021, and is equal to 2 after 2021.

Why would we do this? It could potentially change the way we interpret our results because as time continues, more and more states legalise, yet the number of post-treatment observations shrinks in comparison to earlier cohorts. This matters considering we seeing treatment effects

starting in year 6 post legalisation. In the above scenario, New York does not have 6 years to observe after legalisation, thus making it obsolete so we would want to account for that:



The chart shows us that very early cohorts (2012-2015) have less than 5 active treatments, and it is not until 2019 that we reach 10 active treatments. This poses a couple problems for me to solve. The first is that the post-period results from our double interaction term model are almost entirely a result of the states that legalised within the first 4 years as those are the only states with a post-period that can span beyond 6 years after treatment. As such, the robustness in our model is degraded due to a lack of observations - the estimates are more sensitive to slight differences in the later years. We also cannot assume that these upward trends can apply to the later cohorts (2017-2022).

The best way to deal with this issue is to face it head-on and to restrict our data in the final model to only include states that legalised between 2012 and 2016. These states are: Colorado (2012), Washington (2012), Alaska (2014), Oregon (2014), California (2016), Maine (2016), Massachusetts (2016), and Nevada (2016). Restricting the model to only include these states may decrease the model's confidence in the estimates, however the trade-off is worth it in order to strive to see defend-able results.

It should be noted that after restricting the data like this, the weights from the prior model will be shifted, but this is not an issue as the control group will now be compared to a smaller group of states with more data, meaning the control group will be even more representative of the (new) treated group.

Model 4: Employment Growth Rate Deviations

To further our validity/robustness checks, we now move away from transforming the explanatory variables. Now, we will introduce a new option for the response variable, not for the purpose of including in our final model but rather purely to act as a sanity check that the results we are seeing are not happening by chance. At this point in the report, this may feel redundant, however it is important to explore all possibilities in case a specific model (implemented without error) breaks all of your conclusions.

In this specific instance, I am going to create the variable **RelativeGrowth**. This will be a measure of deviation in employment growth rate across years from the mean national employment growth rate. In other words, it is a measure of how state-specific employment growth compare to the national growth. This will be useful as a response variable to give us a standardised metric by which we can better understand if the states actually experienced significant effects in employment.

Imagine a scenario where the USA is in a recession, say 2019/2020 (COVID). During this time, every state is suffering in terms of employment, but if the national percent change is -10% (for example), but, say, Oregon has only a -4% change in employment. Under this new response variable, Oregon's observation is written as +6%. From this, we could reason that while yes, employment overall was falling, but Oregon was suffering LESS and that difference is a result of the treatment event.

NOTE: These are not percentages. They are the decimal forms of the percent change

1. For each state s in year t , calculate the year-on-year employment growth rate:

$$\text{StateGrowth}_{st} = \frac{\text{Employ}_{st} - \text{Employ}_{s,t-1}}{\text{Employ}_{s,t-1}}$$

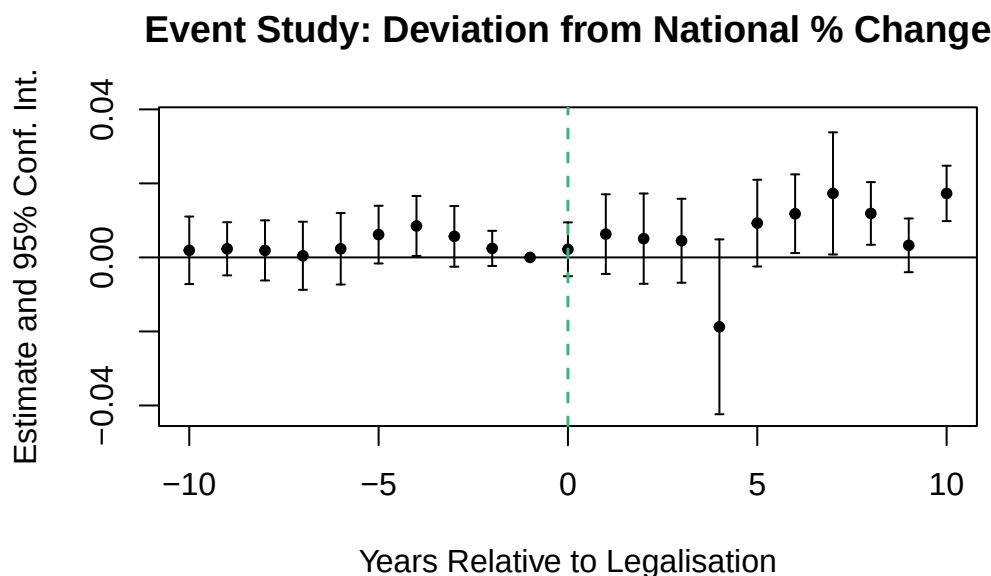
2. Calculate the mean employment growth rate, $\overline{\text{NationalGrowth}_t}$, across all n states for each year t . A positive value means a state grew faster than the national average that year; a negative value means it grew slower.

$$\overline{\text{NationalGrowth}_t} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\text{Employ}_{i,t} - \text{Employ}_{i,t-1}}{\text{Employ}_{i,t-1}} \right)$$

3. Determine the $\text{RelativeGrowth}_{st}$ for each state by subtracting the national average from the individual state's growth rate.

$$\text{RelativeGrowth}_{st} = \text{StateGrowth}_{st} - \overline{\text{NationalGrowth}_t}$$

This approach will be useful in drawing comparative conclusions that are not affected by national trends. For this study, that means if there is any treatment effect observed it is sensible to argue it is causally related to the treatment event. Moreover, it means if we see delayed treatment effects with this variable as the response variable, we can be further confident in our claim that legalising recreational marijuana leads to an increase in employment levels after a brief period of development (the gestation period if you will).



So what do we get? Another result consistent with the rest of our models! There appears to be a treatment effect at approximately $t=6$ which aligns with exactly what the other models have told us insofar. The pre-treatment coefficients are slightly less in agreement with the parallel trends assumption, but this is likely be a result of how we calculated the `GrowthRate` variable. There seems to be a slight cyclical pattern too, however at $t=6$, the cycle is slightly higher than the prior 2(ish) cycles. It would be interesting to observe another cycle after $t=10$, however this is not possible at the moment due to the new-ness of the data.

This response variable will not replace $\ln(\text{Employment})$ in our final model, however it is comforting to know that when using $\ln(\text{Employment})$ we do not have to be concerned about only capturing national trends instead of state trends.

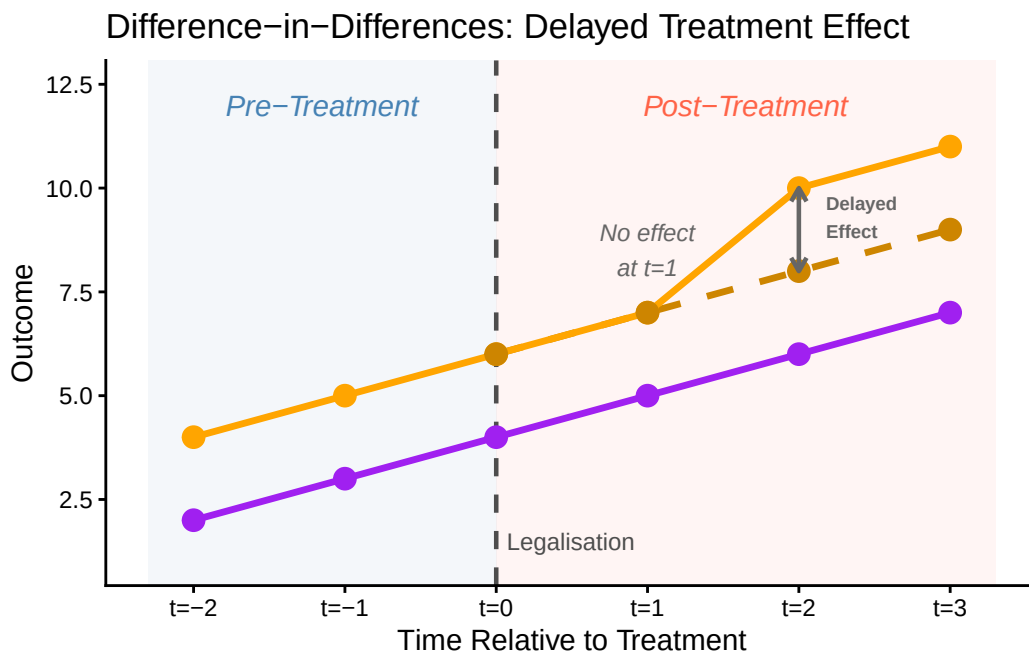
Delayed Treatment Effects

So there are all these models, and they all tell us that we have something called “delayed treatment effects”... What does that really mean?

In simple terms, it means that what we observe is no effect immediately following the treatment event, but rather an effect further along in time. In our case, this “further in time” aspect means roughly 6-7 years after the legislation is passed in a state allowing recreational use of marijuana. During the time between the treatment effect and the observed treatment effect though, it is not entirely accurate to say nothing is happening as a direct consequence of the legislation. It is during this period that the industry is being built. This may include distribution, production, etc licensing along with the creation of designated plantations and store fronts, and/or officially registering the company (and getting the certificate to prove as such). These are all crucial steps given the nature of the treatment in question, however none of them happen overnight. Furthermore, even illegal marijuana businesses need to go through the licensing steps, which often include tests on product to ensure regulations are met (usually defined within state specific legislature).

In other words, “delayed treatment effects” is just a fancy way to say that there is an effect as a result of the treatment event, but it doesn’t happen for a while because other hurdles need to be cleared first. And of course, some businesses will see an almost immediate transition depending on their state’s laws, however because our study uses aggregated data across ALL treated states, these few exceptions are not strong enough to force the results into being significant.

The following graph is an example of this for a simple visual representation of everything mentioned above. Note that this is an example only and is not representative of the actual data within our study, only the overarching final conclusions.



Final Model

Now that I have explored different modelling options, witnessed consistent observation trends, and better understand the data we are working with, I can develop a final model that encapsulates all of the key findings from the previous models.

This final model will include the following features:

- Double Interaction Staggered DiD
- Inverse Probability Weighting
- Early Legalising States ONLY: Colorado, Washington, Alaska, Oregon, California, Maine, Massachusetts, and Nevada

These features were chosen due to the justifications discussed from each model and are believed to be the best way to account for the nuances within the data. I have not made any changes to the control variables and the outcome variable remains as original: $\ln(\text{Employment})$.

We will now look at the regression outputs:

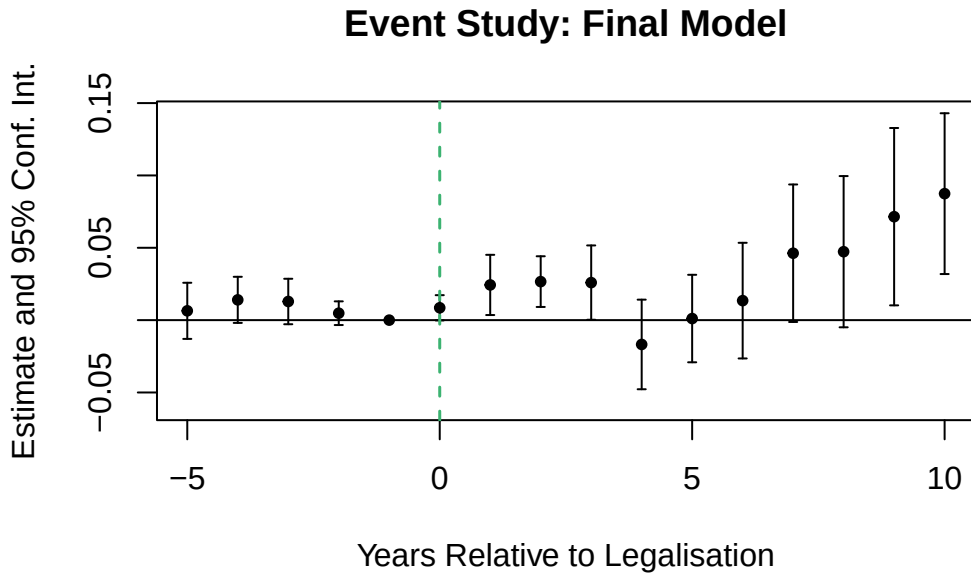
Table 4: Final Model: IPW-Weighted Early Cohort DiD

	T>6	T>7	T>8
During	0.0031 (0.0162)	0.0052 (0.0163)	0.0072 (0.0166)
Post	0.0467+ (0.0255)	0.0519* (0.0250)	0.0672** (0.0218)
log(GDP)	0.0722** (0.0240)	0.0715** (0.0240)	0.0710** (0.0241)
log(Median Income)	0.0031 (0.0027)	0.0029 (0.0026)	0.0029 (0.0027)
Population (Millions)	0.0185** (0.0051)	0.0185** (0.0052)	0.0184** (0.0053)
Pct Male (z)	-0.0447* (0.0210)	-0.0441* (0.0210)	-0.0437* (0.0212)
Pct Black (z)	0.0711 (0.1003)	0.0661 (0.0994)	0.0599 (0.0983)
Num.Obs.	592	592	592
R2	0.999	0.999	0.999
R2 Adj.	0.999	0.999	0.999
R2 Within	0.344	0.341	0.338
R2 Within Adj.	0.335	0.332	0.329
Std.Errors	by: State	by: State	by: State
FE: State	X	X	X
FE: Year	X	X	X
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

This regression output table further proves our belief of delayed treatment effects. It does so because of a couple of reasons:

1. The “During” period is consistently statistically insignificant
2. the “Post” period continually increases in significance and magnitude when I increase the number of years post-legalisation included within “During” - meaning the later the year since treatment, the stronger the effect is felt.

To be sure of the output, let’s visualise the coefficients on an event study one last time:



What we see is an event study that does in fact agree with our regression table outputs. The “Pre” period shows evidence of parallel trends between the control and treatment groups, meaning the IPW weighting was successful. We then see delayed treatment effects approximately 7 years post-legalisation.

After observing consistent results between the final model and the individual specification models, we can confidently make a causal argument that the legalisation of recreational marijuana does indeed cause a delayed treatment effect, however we do need to acknowledge limitations and concerns that still remain. This brings me to my concluding thoughts.

Conclusion

The goal of this report was to investigate the impact of legalising recreational marijuana on US state employment levels via an approach based in the foundations of a Difference-In-Differences model. In this, the research proved successful as we do observe a positive, delayed treatment effect. As such, we can tentatively claim a causal relationship in this discipline. That being said, we cannot definitively make this claim. Firstly, the model relies heavily upon the parallel trends assumption, which while we do satisfy, it is impossible for states to actually achieve perfect parallelism. Secondly, given we observe the treatment effect around 6-7 years after legalisation, we are seeing the trends of only a few states that legalised early enough for us to observe that length of time after treatment. It is plausible that other states may not experience equivalent nor similar results and that the states we DO have the data for happened to see

a positive relationship between legalisation and employment simply by chance. Finally, there are always unobserved differences that, no matter the mathematical formula applied, cannot be accounted for. In our model, IPW does a good job at trying to achieve this goal, however it is not a perfect measure.

This project took place across the course of a 10-week period and as a result, there is a lot of room for improvement if this study were to be continued. Options for further study include:

- Incorporating a double treatment model to include medical marijuana legalisation effects.
- Comparing the US's results to other countries that have partially or totally legalised marijuana (such as Canada, Uruguay, Netherlands). This would serve as an interesting comparison to determine if the effects observed within the USA are US-specific or are a general result of marijuana.
- Including industry data to determine if there is a difference across sectors of the labour market instead of a general overview, similar to the methodology of Varon. This would open the door to clustering by industry also which could yield further interesting conclusions.

Acknowledgements

None of this report would be truly possible with myself alone. Henceforth, I would like to take the opportunity to thank my fellow classmates in ECON602: Senior Research Experience (Winter 2026) for the guidance, feedback, engagement, and honesty throughout the course of the term. Furthermore, I would like to extend another heartfelt thank you to Professor Lhost for really helping to shape this project into what it is now, and directing me on paths that led to meaningful insights and development of myself as a researcher.

Moreover, in the modern age it is undeniable that I am not alone in creating the visualisations used within this report. As such, I would like to acknowledge that AI was used strictly for the following purposes: To produce code for the example DiD models, to assist with event study graph aesthetics including text placement, shading, and axis re-scaling, and to help clarify syntax for the `iplot` and `modelsummary` packages as well as LaTeX formatting for model display in my rendered PDF output. This being said, all underlying analytical decisions, model choices, mutations, and interpretations stem from either my own original thought or from in-class discussion. Finally, AI was consulted to resolve an error in the rendered PDF in which the regression output tables were printing in the wrong order as compared to where it was specified within the `.qmd` file.

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